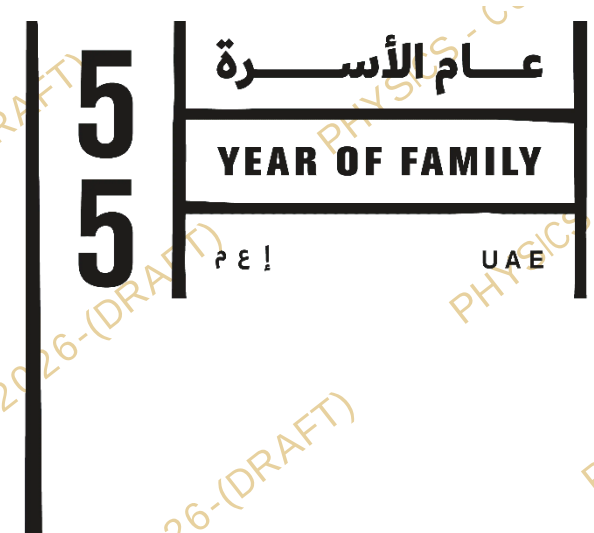
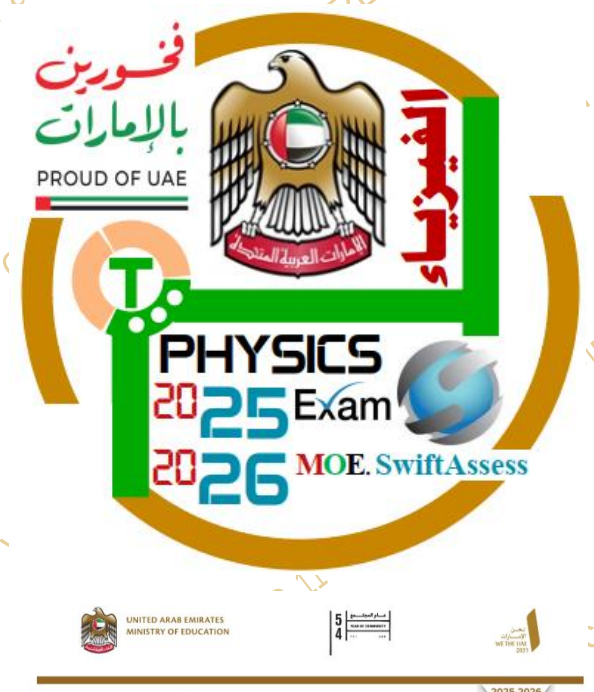


Academic Year العام الدراسي	2K ²⁵ /2K ²⁶
Term/الفصل	3
Subject الموضوع	PHYSICS (INSPIRE)
Grade الصف	11
Stream. المسار	Advanced المتقدم
Number Of MCQ عدد الأسئلة الموضوعية	25
Markes of MCQ درجة الأسئلة الموضوعية	4
Type of All Questions نوع كافة الأسئلة	MCQ الأسئلة الموضوعية
Maximum Overall Grade الدرجة القصوى الممكنة	100
Exam Duration مدة الامتحان	120 min
Mode of implementation طريقة التطبيق	Swift Assess.
Calculator الآلة الحاسبة	Allowed مسموحة



Type النوع	Question السؤال *	ناتج التعلم/ معايير الأداء** Learning Outcome/Performance Criteria**	المرجع في كتاب الطالب (النسخة الانجليزية) Reference in the Student Book (English Version)	
			Example/Exercise. مثال/تمرين	Page. صفحة
الأسئلة الموضوعية [100% - MCQ]	1	- Define a radian and convert degrees to radians and vice-versa. - Identify that counterclockwise motion is in the positive direction and clockwise motion is in the negative direction.	Student Book Get It?	66~67 67
	2	Relate the arc length (x) to the angular displacement (θ) and the distance (r) from the axis of rotation.	Student Book	67
	3	- Apply the relationship between average angular velocity (ω), angular displacement ($\Delta\theta$), and the time interval (Δt) for that displacement. - Solve problems related to rotational variables.	Student Book PRACTICE Pro. Q (2) Check Your Prog. Q (6,7)	67~68 69 69
	4	- Relate the linear velocity (v) to the angular velocity (ω) and the distance (r) from the axis of rotation. - Solve problems related to rotational variables.	Student Book Check Your Prog. Q (10)	68 69
	5	- Apply the relationship between average angular acceleration (α), change in angular velocity ($\Delta\omega$), and the time interval (Δt) for that change. - Solve problems related to rotational variables.	Student Book Check Your Prog. Q (8)	68 69
	6	- Define the term torque. - Identify that a torque on a body involves a force and a lever arm which is the perpendicular distance from the axis of rotation to the point where the force is applied.	Student Book Figure. (5) & Figure. (6)	70~71 70,71
	7	- Apply the relation ($\tau = Fr\sin\theta$) to calculate the magnitude of torque (τ) for a force of magnitude (F) where (r) is the distance from the axis of rotation to the point where the force is exerted, and θ is the angle between the force and the radius from the axis of rotation to the point where the force is applied. - Identify that a torque is assigned a positive or negative sign depending on the direction it tends to make the body rotate about a specified rotation axis: "clocks are negative"	Student Book EXAMPLE Pro. Q (1) PRACTICE Pro. Q (11,12,13)	71 72 72
	8	- Solve problems that involve torque. - Calculate the net torque when more than one torque acts on a body about a rotation axis.	Student Book EXAMPLE Pro. Q (2) PRACTICE Pro. Q (16,18)	73 75 75
	9	Define moment of inertia as the resistance to rotation.	Student Book	75
	10	Calculate the moment of inertia (rotational inertia) of a single point mass rotating about an axis ($I=mr^2$).	EXAMPLE Pro. Q (3) PRACTICE Pro. Q (21)	77 78
	11	- Demonstrate how the moment of inertia of an extended object depends on the location and direction of the rotational axis. - Determine the moment of inertia of extended objects like, solid uniform cylinder, uniform sphere, long uniform rod.	Student Book Table 2	76 76
	12	- State newton's second law for rotational motion which relates the net torque (τ_{net}) on a body to the body's moment of inertia (I) and rotational acceleration (α), all calculated relative to a specified axis of rotation. - Apply newton's second law for rotation to solve problems.	Student Book EXAMPLE Pro. Q (4) Check Your Prog. Q (35)	78 79 80
	13	- Explain the meaning of center of mass. - Locate experimentally the center of mass of an object by suspending it from different points.	Student Book Check Your Prog. Q (42,44,45)	81~83 87
	14	State the conditions for an object to be in static equilibrium- both translational and rotational equilibrium.	Student Book Get It? Check Your Prog. Q (41)	83 86 87
	15	Explain what happens when a player hits a ball in terms of the acting forces and their effects on the colliding objects before, during, and after collision.	Student Book	93
	16	- Define impulse as the product of average force on an object and the time interval over which it acts. - Derive impulse expression from Newton's Second Law of motion and specify its unit.	Student Book	93
	17	Calculate the impulse in case of a <u>constant force</u> and in case when the <u>force is not constant</u> by using the average force multiplied by the time interval over which it acts or by finding the area under a force-time graph.	Student Book PRACTICE Pro. Q (3,4)	93~94 95
	18	Apply the linear momentum equation to calculate the linear momentum of different systems.	Student Book PRACTICE Pro. Q (1)	94
	19	Apply the impulse-momentum theorem to solve relevant problems	Student Book EXAMPLE Pro. Q (1) PRACTICE Pro. Q (2)	94 95 96
	20	Define the angular momentum of an object as the product of the object's moment of inertia and the object's angular velocity and write in in equation form ($L=I\omega$) and specify its unit.	Student Book	97
	21	- Solve problems involving angular momentum. - Solve problems involving the conservation of angular momentum.	EXAMPLE Pro. Q (2)	99
	22	Discuss the forces and impulses exerted on one another in case of two-particle collision in one dimension.	Student Book Check Your Prog. Q (30,32)	101 110
	23	- Define a closed system as a system that does not gain or lose mass. - Define an isolated system as a closed system where the net of external forces is zero.	Student Book	102
	24	Define the system and apply the law of conservation of linear momentum to different situations in one dimension like recoil, propulsion in space, or others to calculate different physical quantities.	Student Book EXAMPLE Pro. Q (3) EXAMPLE Pro. Q (4)	102~105 103 105
	25	- Explain the law of conservation of angular momentum and express it in equation form ($L_f=L_i$) - Discuss different examples that involve the law of conservation of angular momentum like in ice skating, tops, gyroscopes, and others. - Solve problems involving the conservation of angular momentum.	Student Book Figure. (13)	108 108
* قد تظهر الأسئلة بترتيب مختلف في الامتحان الفعلي. / Questions might appear in a different order in the actual exam.				
** كما وردت في كتاب الطالب وLMS والخطة الفصلية. / As it appears in the textbook, LMS, and (Main_IP).				